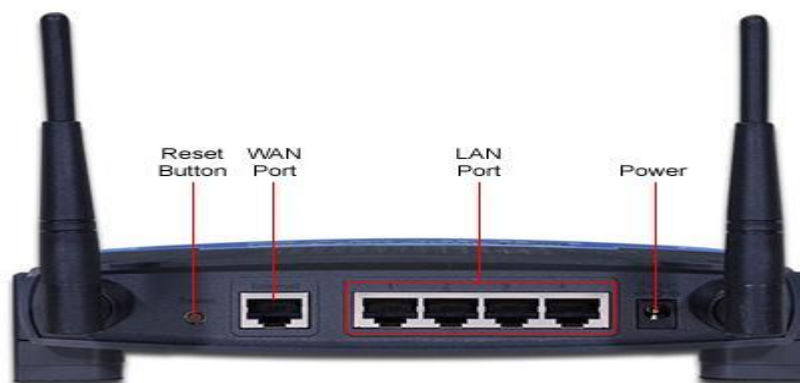


1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home Wi-Fi setup.

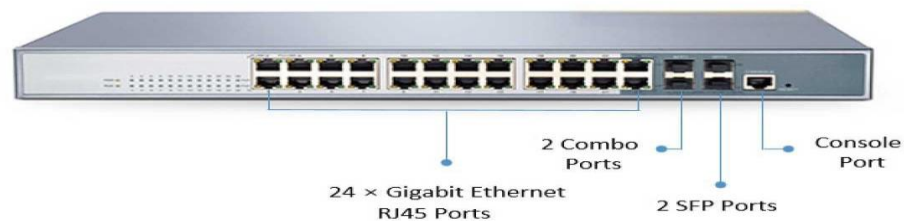
Network Device	Main Function	Typical Ports
Switch	Connects multiple wired devices within the same local area network and forward data to the correct device using MAC address.	5,8,16,24 or 48 Ethernet ports (10/100/1000 Mbps or higher)
Router	Connect the home network to the internet, assigns IP addresses and performs Network Address Translation.	1 WAN port and 2 to 8 LAN Ethernet port
hub	Connect multiple wired devices but broadcasts incoming data to all connected devices without filtering or switching.	Usually 4, 8 or 16 Ethernet (RJ-45) ports
Firewall	Monitors and filters incoming and outgoing network traffic to protect the network from unauthorised access and cyber threats.	Often 2 to 8 Ethernet (RJ-45) ports
Access Point	Provides Wi-Fi connectivity to devices and extends or creates a wireless network connected to the wired LAN.	1 or 2 Ethernet port

2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, and antenna) for at least three devices.



Labels

- Device : Wireless router
- Ethernet (LAN) ports: Connect PCs, switches, printers, etc.
- Ethernet (WAN/Internet) port: Connects to the ISP.
- Power port: Connects the power adapter.
- Antennas: Provides Wi-Fi signal.



Labels

- Device : Switch
- Ports/components :
- Ethernet Ports (RJ-45) – Connect computers, printers, access points and other devices.
- Console Port – Used for configuration and management.
- Power Port – Supplies electrical power to the switch.



Labels

- Device : Wireless Access Point
- Ports/Components:
- Ethernet Port – Connects the access point to the network.
- Power Port – Used if PoE is not available.
- Antennas – Transmits and receive wireless signals.

3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

- **Example of Instagram**
- When I open Instagram, my phone first connects to the home Wi-Fi Access Point, which provides a wireless connection to the network. The data is then sent to a Router, which forwards the internet traffic to Instagram's servers. A Firewall helps protect the network by filtering unwanted or malicious traffic; while Switches inside the internet service providers and data centre's networks efficiently transfer data between devices and servers.

4. Suppose your college Wi-Fi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

- **Possible reason:**
- The college has too many users connected to a single Wi-Fi access point. When the access point reaches its maximum capacity, it becomes overloaded, causing slow internet speeds and frequent disconnections, especially during busy hours.
- **How upgrading one device could help:**
- Upgrading to a higher-capacity, modern wireless access point would allow it to handle more connected devices simultaneously, improve wireless coverage, reduce network congestion and provide faster, more stable connections for students and staff.
- A possible reason for the slow and unstable college Wi-Fi is that the wireless access point is overloaded with too many connected users. This causes network congestion,

resulting in slow speeds and frequent disconnections. Upgrading to a higher-capacity Wi-Fi 6 access point would improve coverage, support more devices at the same time, and provide a faster, more reliable wireless connection.